



TITLE	: WATER DEMAND
PROJECT NAME	: PROPOSED RESIDENTIAL & COMM. AT PUNAVALE
CLIENT NAME	: MR.PRATHMESH GAIKWAD
PROJECT LOCATION	: PUNAVALE, PUNE
PROJECT NO.	: 078

[illegible]

DESIGN ENGINEER NAME :- MR.AKASH KHADE

DATE :- 30-07-2026

CONSOLIDATED STATEMENT									
SR.NO	DISCRIPTION	PLOT - A			PLOT - B			TOTAL	UNITS
		RESIDENTIAL	COMM	SUB-TOTAL	RESIDENTIAL	COMM	SUB-TOTAL		
SECTION-7									
1	Number Of Building	4	2	6	4	1	5	11	NO.S
2	Total Number Of Flats	294	-	294	224	-	224	518	NO.S
3	Total Number Of Residential Building Population	1470	964	2434	1120	113	1233	3667	NO.S
SECTION-8									
DRY SEASON									
1	Fresh Water Requirement	154.35	33.49	187.84	117.60	2.83	120.43	308.27	KLD
2	Flushing Water Requirement	44.10	17.72	61.82	33.60	2.26	35.86	97.68	KLD
3	Landscape Water Requirement	4.59	0.00	4.59	4.57	0.00	4.57	9.16	KLD
4	Swimming Pool Makeup Water Requirement	0.00	0.00	0.00	0.00	0.00	0.00	0.00	KLD
5	HVAC Water Requirement	0.00	0.00	0.00	0.00	0.00	0.00	0.00	KLD
6	Total Water Requirement	203.04	51.21	254.25	155.77	5.09	160.86	415.10	KLD
7	Total Treated Water	158.76	40.96	199.72	125.03	0.00	125.03	324.75	KLD
8	Excess Treated Water To Corporation Line	110.07	23.25	133.32	59.77	0.00	59.77	193.09	KLD
WET SEASON									
1	FRESH WATER REQUIREMENT	154.35	33.49	187.84	117.60	2.83	120.43	308.27	KLD
2	FLUSHING WATER REQUIREMENTS	44.10	17.72	61.82	33.60	2.26	35.86	97.68	KLD
3	LANDSCAPE WATER REQUIRED	0.00	1.00	2.00	1.00	0.00	1.00	3.00	KLD
4	SWIMMING POOL MAKEUP WATER REQUIRMENT	0.00	0.00	0.00	0.00	0.00	0.00	0.00	KLD
5	HVAC WATER REQUIREMENT	0.00	0.00	0.00	0.00	0.00	0.00	0.00	KLD
6	TOTAL WATER REQUIREMENT	198.45	52.21	251.66	152.20	5.09	157.29	408.94	KLD
7	TOTAL TREATED WATER	158.76	40.96	199.72	125.03	0.00	125.03	324.75	KLD
8	EXCESS TREATED WATER WATER TO COPORATION LINE	114.66	23.25	137.91	64.34	0.00	64.34	202.25	KLD
SECTION-9									
UGT DETAILS									
1	DOMESTIC UGT CAPACITY	309.00	68.00	377.00	236.00	6.00	242.00	619.00	LIT/DAY
2	FLUSHING UGT CAPACITY	45.00	18.00	63.00	34.00	3.00	37.00	100.00	LIT/DAY
3	FIRE UGT CAPACITY	300.00	260.00	560.00	230.00	0.00	230.00	790.00	LIT/DAY
SECTION-10									
STP DETAILS									
1	SEWAGE GENERATION	178.61	46.08	224.69	140.66	0.00	140.66	365.35	LIT/DAY
2	STP Capacity	180.00	50.00	230.00	150.00	0.00	150.00	380	LIT/DAY

DESIGN ENGINEER NAME :- MR.AKASH KHADE											
DATE :- 30-07-2026											
WATER DEMAND											
PLOT - A											
SECTION-1		RESIDENTIAL									
SR.NO	BLDG/ WING	NO. OF FLAT	POPULATION PER FLAT	POPULATION	DOMESTIC WATER DEMAND (LIT/DAY)		FLUSHING WATER DEMAND (LIT/DAY)		TOTAL WATER DEMAND (LIT/DAY)		
1	WING - A	146	5	730	105	76650	30	21900	135	98550	
2	WING - B	146	5	730	105	76650	30	21900	135	98550	
3	BUNGLOW-A	1	5	5	105	525	30	150	135	675	
4	BUNGLOW-B	1	5	5	105	525	30	150	135	675	
										198450	
SUB-TOTAL		294		1470		154350		44100		198450	
SECTION-2		WATER REQUIRMENTS FOR OTHER MEASURES									
1	WATER REQUIREMENT FOR LANDSCAPE-PLOT-A (AS PER NBC-2026)			AREA (SQ.M)	WATER REQUIREMENT	UNITS					
				765	4590	LITER/DAY					
2	MAKE UP WATER FOR SWIMMING POOL				0	0					LITER/DAY
3	WATER REQUIREMENT FOR HVAC					0					LITER/DAY
SECTION-3		COMMERCIAL									
SR.NO	BLDG/ WING	AREA (SQ.M)	POPULATION PER/SQ.M	POPULATION	DOMESTIC WATER DEMAND (LIT/DAY)		FLUSHING WATER DEMAND (LIT/DAY)		TOTAL WATER DEMAND (LIT/DAY)		
COMM-A											
1	Ground Floor (Shop)	437	3	146	25	3650	20	2920	45	6570	
2	1st Floor (Shop)	437	6	73	25	1825	20	1460	45	3285	
3	2nd to 6th (Office)	2185	10	219	25	5475	20	4380	45	9855	
4	7th (Resturant)	437	1.4	313	55	17215	15	4695	70	21910	
										41620	
SUB-TOTAL				751		28165		13455		41620	

COMM-B										
1	Gr.+Mezz Floor (Shop)	213	3	71	25	1775	20	1420	45	3195
2	1st & 2nd (office)	484	10	49	25	1225	20	980	45	2205
3	3rd & 4Th (office)	924	10	93	25	2325	20	1860	45	4185
										9585
SUB-TOTAL				213		5325		4260		9585
GRAND TOTAL RESIDETIAL + COMMERCIAL		294		2434		187840		61815		249655
PLOT - B										
SECTION-4		RESIDENTIAL								
SR.NO	BLDG/ WING	NO. OF FLAT	POPULATION PER FLAT	POPULATION	DOMESTIC WATER DEMAND (LIT/DAY)		FLUSHING WATER DEMAND (LIT/DAY)		TOTAL WATER DEMAND (LIT/DAY)	
1	WING - C	56	5	280	105	29400	30	8400	135	37800
2	WING - D	56	5	280	105	29400	30	8400	135	37800
3	WING - E	56	5	280	105	29400	30	8400	135	37800
4	WING - F	56	5	280	105	29400	30	8400	135	37800
										151200
SUB-TOTAL		224		1120		117600		33600		151200
SECTION-5		WATER REQUIRMENTS FOR OTHER MEASURES								
1	WATER REQUIREMT FOR LANDSCAPE (AS PER NBC-2026)				AREA (SQ.M)	WATER REQUIREMENT	UNITS			
					762	4572	LITER/DAY			
2	MAKE UP WATER FOR SWIMMING POOL				0	0	LITER/DAY			
3	WATER REQUIREMT FOR HVAC					0	LITER/DAY			
SECTION-6		COMMERCIAL								
SR.NO	BLDG/ WING	AREA (SQ.M)	POPULATION PER/SQ.M	POPULATION	DOMESTIC WATER DEMAND (LIT/DAY)		FLUSHING WATER DEMAND (LIT/DAY)		TOTAL WATER DEMAND (LIT/DAY)	
COMM-C										
1	Ground & Mezz	339	3	113	25	2825	20	2260	45	5085
										5085
SUB-TOTAL				113		2825		2260		5085
GRAND TOTAL RESIDETIAL + COMMERCIAL		224		1233		120425		35860		156285

NOTE :- WATER DEMAND CALCULATION AS PER THE NBCS-2026

SECTION-7					
*UGT & OHT DETAILS					
PLOT -A					
1	FOR RESIDENTIAL				
SR.NO.	DISCRIPTION	WATER REQUIREMENT (LIT/DAY)	STORAGE OF WATER (DAYS)	TOTAL WATER STORAGE (LIT/DAY)	TOTAL WATER STORAGE (KLD)
1	DOMESTIC WATER TANK	154350	2	309000	309.00
2	FLUSHING WATER TANK NEAR STP	44100	1	45000	45.00
3	FIRE WATER TANK	300000	1	300000	300.00
TOTAL STORAGE CAPACITY					654.00
NOTE:- FIRE WATER TANK CAPACITY TAKEN AS PER NBCS-2026, SO KINDLY CONFIRM WITH FIRE LIOSANING VENDOR & NOC					
2	FOR COMMERCIAL-A				
SR.NO.	DISCRIPTION	WATER REQUIREMENT (LIT/DAY)	STORAGE OF WATER (DAYS)	TOTAL WATER STORAGE (LIT/DAY)	TOTAL WATER STORAGE (KLD)
1	DOMESTIC WATER TANK	28165	2	57000	57.00
2	FLUSHING WATER TANK (A&B)	17715	1	18000	18.00
3	FIRE WATER TANK	230000	1	230000	230.00
TOTAL STORAGE CAPACITY					305.00
3	FOR COMMERCIAL-B				
SR.NO.	DISCRIPTION	WATER REQUIREMENT (LIT/DAY)	STORAGE OF WATER (DAYS)	TOTAL WATER STORAGE (LIT/DAY)	TOTAL WATER STORAGE (KLD)
1	DOMESTIC WATER TANK	5325	2	11000	11.00
2	FLUSHING WATER TANK	0	1	0	0.00
3	FIRE WATER TANK	30000	1	30000	30.00
TOTAL STORAGE CAPACITY					41.00
*OHT DETAILS					
SR.NO	BLDG/ WING	DOMESTIC (KLD)	FLUSHING (KLD)	FIRE BREAK TANK (KLD)	FIRE OHT TANK (KLD)
RESIDENTIAL & COMMERCIAL					
1	WING - A	77.00	22.00	100.00	100.00
2	WING - B	77.00	22.00	100.00	100.00
3	BUNGLOW-A	1.00	1.00	0.00	0.00
4	BUNGLOW-B	1.00	1.00	0.00	0.00
5	COMMERCIAL-A	29.00	14.00	0.00	20.00
6	COMMERCIAL-B	6.00	5.00	0.00	20.00
TOTAL OHT CAPACITY		154.00	44.00	200.00	200.00
NOTE:- FIRE WATER TANK CAPACITY TAKEN AS PER NBC-2026, SO KINDLY CONFIRM WITH FIRE LIOSANING VENDOR & NOC					
SECTION-8					

PLOT -B					
1	FOR RESIDENTIAL				
SR.NO.	DISCRIPTION	WATER REQUIREMENT (LIT/DAY)	STORAGE OF WATER (DAYS)	TOTAL WATER STORAGE (LIT/DAY)	TOTAL WATER STORAGE (KLD)
1	DOMESTIC WATER TANK	117600	2	236000	236.00
2	FLUSHING WATER TANK NEAR STP	33600	1	34000	34.00
3	FIRE WATER TANK	230000	1	230000	230.00
TOTAL STORAGE CAPACITY					500.00
NOTE:- FIRE WATER TANK CAPACITY TAKEN AS PER NBCS-2026, SO KINDLY CONFIRM WITH FIRE LIOSANING VENDOR & NOC					
2	FOR COMMERCIAL-C				
SR.NO.	DISCRIPTION	WATER REQUIREMENT (LIT/DAY)	STORAGE OF WATER (DAYS)	TOTAL WATER STORAGE (LIT/DAY)	TOTAL WATER STORAGE (KLD)
1	DOMESTIC WATER TANK	2825	2	6000	6.00
2	FLUSHING WATER TANK	2260	1	3000	3.00
3	FIRE WATER TANK	0	1	0	0.00
TOTAL STORAGE CAPACITY					9.00
*OHT DETAILS					
SR.NO	BLDG/ WING	DOMESTIC (KLD)	FLUSHING (KLD)	FIRE BREAK TANK (KLD)	FIRE OHT TANK (KLD)
RESIDENTIAL & COMMERCIAL					
1	WING - C	30.00	9.00	0.00	20.00
2	WING - D	30.00	9.00	0.00	20.00
3	WING - E	30.00	9.00	0.00	20.00
4	WING - F	30.00	9.00	0.00	20.00
5	COMMERCIAL - C	3.00	3.00	0.00	0.00
TOTAL OHT CAPACITY		60.00	18.00	0.00	60.00
NOTE:- FIRE WATER TANK CAPACITY TAKEN AS PER NBC-2026, SO KINDLY CONFIRM WITH FIRE LIOSANING VENDOR & NOC					

SECTION-9			
STP DETAILS PLOT-A			
1	STP FOR RESIDENTIAL		
SR.NO.	DISCRIPTION	CAPACITY	UNITS
1	TOTAL WATER REQUIREMENT FOR RESIDENTIAL TENAMENTS	198450	LITERS/DAY
2	TOTAL SEWAGE GENERATION @90% REQUIREMENT (10% INFLTRATION LOSS)	178605	LITERS/DAY
3	CAPACITY OF SEWAGE GENERATION	178.61	KLD
4	SAY STP CAPACITY	180.00	KLD
5	TREATED WATER AFTER FILTRATION STP PROCESS	158760	LITERS/DAY
6	REUSE WATER FOR FLUSHING	44100	LITERS/DAY
7	REUSE WATER FOR LANDSCAPE	4590	LITERS/DAY
8	REUSE WATER FOR HVAC	0	LITERS/DAY
9	EXCESS TREATED WATER TO EXTERNAL MUNICIPAL DRAIN	110070	LITERS/DAY

1	STP FOR COMMERCIAL A & B		
SR.NO.	DISCRIPTION	CAPACITY	UNITS
1	TOTAL WATER REQUIREMENT FOR RESIDENTIAL TENAMENTS	51205	LITERS/DAY
2	TOTAL SEWAGE GENERATION @90% REQUIREMENT (10% INFLTRATION LOSS)	46085	LITERS/DAY
3	CAPACITY OF SEWAGE GENERATION	46.08	KLD
4	SAY STP CAPACITY	50.00	KLD
5	TREATED WATER AFTER FILTRATION STP PROCESS	40964	LITERS/DAY
6	REUSE WATER FOR FLUSHING	17715	LITERS/DAY
7	REUSE WATER FOR LANDSCAPE	0	LITERS/DAY
8	REUSE WATER FOR HVAC	0	LITERS/DAY
9	EXCESS TREATED WATER TO EXTERNAL MUNICIPAL DRAIN	23249	LITERS/DAY

STP DETAILS PLOT-B			
1	STP FOR RESIDENTIAL & COMMERCIAL		
SR.NO.	DISCRIPTION	CAPACITY	UNITS
1	TOTAL WATER REQUIREMENT FOR RESIDENTIAL TENAMENTS	156285	LITERS/DAY
2	TOTAL SEWAGE GENERATION @90% REQUIREMENT (10% INFILTRATION LOSS)	140657	LITERS/DAY
3	CAPACITY OF SEWAGE GENERATION	140.66	KLD
4	SAY STP CAPACITY	150.00	KLD
5	TREATED WATER AFTER FILTRATION STP PROCESS	125028	LITERS/DAY
6	REUSE WATER FOR FLUSHING	35860	LITERS/DAY
7	REUSE WATER FOR LANDSCAPE	29400	LITERS/DAY
8	REUSE WATER FOR HVAC	0	LITERS/DAY
9	EXCESS TREATED WATER TO EXTERNAL MUNICIPAL DRAIN	59768	LITERS/DAY